

Cloud Computing Basics

Cloud computing is the delivery of computing services over the internet. These services include servers, storage, databases, networking, software, and analytics.

Types of Cloud Services

Infrastructure as a Service (IaaS)

IaaS provides virtualized computing resources over the internet. Users can rent servers, storage, and networking resources instead of purchasing physical hardware.

Examples:

- Amazon EC2
- Microsoft Azure Virtual Machines
- Google Compute Engine

Platform as a Service (PaaS)

PaaS provides a platform for developers to build, test, and deploy applications without managing the underlying infrastructure.

Examples:

- Google App Engine
- Heroku
- Azure App Service

Software as a Service (SaaS)

SaaS delivers software applications over the internet. Users can access applications through a web browser without installing them locally.

Examples:

- Gmail
- Microsoft 365
- Salesforce

Advantages of Cloud Computing

1. Cost Savings
2. Scalability
3. High Availability
4. Disaster Recovery
5. Global Accessibility

Amazon Web Services (AWS)

AWS is the world's leading cloud computing platform. It offers more than 200 cloud services.

Popular AWS Services:

Amazon EC2

Amazon Elastic Compute Cloud (EC2) provides scalable virtual servers in the cloud.

Amazon S3

Amazon Simple Storage Service (S3) is used for storing and retrieving data.

AWS Lambda

AWS Lambda is a serverless computing service that allows users to run code without managing servers.

Cloud Deployment Models

Public Cloud

Resources are owned and operated by a third-party cloud provider.

Private Cloud

Resources are used exclusively by a single organization.

Hybrid Cloud

A combination of public and private cloud environments.

Conclusion

Cloud computing enables organizations to innovate faster, reduce costs, and scale applications efficiently. AWS remains one of the most widely adopted cloud platforms globally.